

Incorporating Receptor Flexibility into DockThor Using Ensemble Docking Approaches

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Background

Molecular recognition is a fundamental process in biological systems, primarily governed by the **induced fit** and **conformational selection** models^[1] (Fig. 1). Molecular docking is a computational technique that simulates these recognition events by predicting the preferred orientation and binding affinity between a ligand and its biological target^[2]. However, accounting for receptor flexibility remains a major challenge in docking methodologies, even among the most advanced approaches in the field^[3].

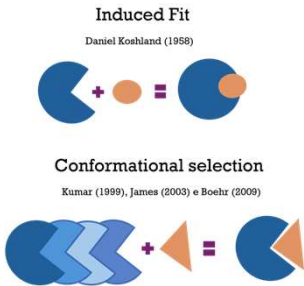


Fig 1. Schematic representation of the two main molecular recognition theories: induced fit and conformational selection.

OBJECTIVES:

- Incorporate receptor flexibility representation into docking experiments using the DockThor program.
- Enhance docking accuracy in DockThor, particularly in cases where protein flexibility plays a critical role.

DockThor Program

DockThor is a molecular docking program entirely developed by Brazilian students and researchers at the Molecular Modeling Group of Biological Systems (GMMSB) at the National Laboratory for Scientific Computing (LNCC/MCTI):

- Freely available at: www.dockthor.lncc.br;
- Developed in C++;
- Utilizes a steady-state genetic algorithm;
- Grid-based docking approach;
- Supports two types of scoring functions:
 - Physics-based.
 - Empirical.

Previous studies

Ensemble docking approach enhances docking accuracy with DockThor in Astex Non-Native Set:

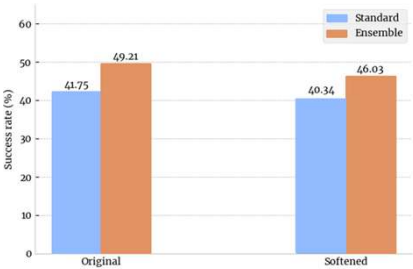
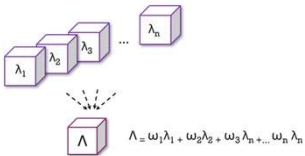


Fig 2. Docking results on the Astex Non-Native Set using ensemble docking to select the best pose per target, evaluated with both versions of the DockThor pose prediction energy function.

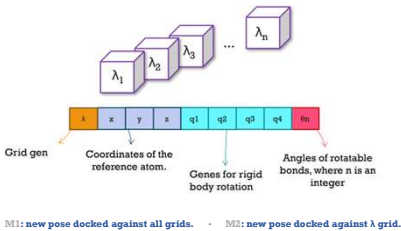
Methodology

Protein flexibility was included using the **ensemble docking** approach, through two complementary strategies:

- Combined Grid (CG):** generates a unified potential grid by linearly combining grids from multiple protein conformations:



- Multiple Grids (MG):** which docks ligands independently against grids from different protein conformations:



Preliminary results

All ensemble strategies either improved or maintained DockThor's accuracy in the preliminary tests (Fig. 4). Using five protein conformations, the computational time increased up to $\sim 3.7\times$ (GM2) compared to single-structure docking (Tab. 1).

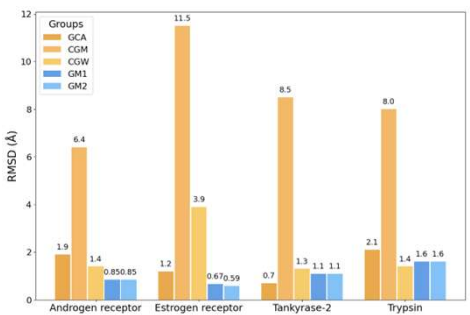


Fig 4. Preliminary tests of MG strategy applications on four selected Core Set targets, presented as median RMSD values for each target based on redocking structures.

Among the CG methodologies, the weighted average strategy achieved the best performance in our tests, whereas the minimum potential grid (CGM) consistently failed.

	Total time	Grid Time	GA Time
Default	30	18	12
CG	70	58	12
GM1	102	58	44
GM2	112	58	54

Tab 1. Execution time (s) of DockThor with the new features for five structures of protein. The 'default' corresponds to DockThor run considering one receptor as a rigid molecule.

HIGHLIGHTS:

- Grid-based ensemble docking allows receptor flexibility with a **good balance** between **accuracy** and **computational cost**.
- The new features will increase DockThor's capability to model molecular recognition more accurately and to generate poses more effectively.
- These enhancements improve the applicability of DockThor in more realistic molecular recognition experiments, reinforcing its value as a tool for drug discovery and computational biology.
- The updated version will soon be available at dockthor.lncc.br.

References

1. Boehr, David D., Ruth Nussinov, and Peter E. Wright. "The role of dynamic conformational ensembles in biomolecular recognition." *Nature chemical biology* 5.11 (2009): 789-796.
2. Salmaso, Veronica, and Stefano Moro. "Bridging molecular docking to molecular dynamics in exploring ligand-protein recognition process: An overview." *Frontiers in pharmacology* 9 (2018): 923.
3. Sahu, Mahendra Kumar, et al. "Exploring recent updates on molecular docking: Types, method, application, limitation & future prospects." *International Journal of Pharmaceutical Research and Allied Sciences* 13.2-2024 (2024): 24-40.
4. Su, Minyi, et al. "Comparative assessment of scoring functions: the CASF-2016 update." *Journal of chemical information and modeling* 59.2 (2019): 895-913.
5. Vendrek, Marcel L., et al. "Protein-ligand docking against non-native protein conformers." *Journal of chemical information and modeling* 48.11 (2008): 2214-2225.